Surgical Instrument Segmentation in Laparoscopic Video

A Literature-Based Framework Combining Multi-Scale Fusion, Refinement, and Temporal Consistency

Antwi-Agyei Nathan Papa
Department of Computer Science, Valley View University
COSC 440: Computer Vision
Instructor: Mr. Sebuabe

First submitted: Friday, 16 January 2026
Revised edition: 13 October 2026

Abstract

Laparoscopic surgery gives the surgical team a two-dimensional video view of the operative field. Computer-assisted systems that can reliably isolate surgical instruments in that view are a foundation for tool tracking, surgical phase recognition, and skill assessment. This paper reviews recent deep-learning work on surgical instrument segmentation and proposes a conceptual segment-then-refine framework that combines multi-scale feature extraction, boundary refinement, and temporal consistency modeling. The review draws on benchmark studies, segment-then-classify and spatiotemporal methods, and recent vision-language-model (VLM) guided refinement, and it asks how each addresses intraoperative challenges such as specular reflection, smoke, blood, and occlusion. The reviewed literature indicates that temporal modeling and multi-scale fusion each improve robustness over single-frame, single-scale baselines, and that surgical context and VLM-based quality assessment are promising but still early-stage additions. This is a literature-based proposal and reports no new experiments; it closes with a concrete validation plan using public laparoscopic datasets. The paper contributes a clear synthesis of the field, a staged design that can be implemented on modest hardware, and an evaluation protocol that treats safety and temporal stability as first-class criteria.

Keywords: surgical instrument segmentation, laparoscopic video, deep learning, medical computer vision, semantic segmentation, temporal consistency

Surgical Instrument Segmentation in Laparoscopic Video

Introduction

Minimally invasive surgery (MIS) has changed modern operative care by reducing trauma and shortening recovery for patients. The cost of these benefits is a more demanding working environment for the surgical team, who must operate from a two-dimensional video feed provided by a laparoscope. Within this restricted view, a core task for computer vision is the precise identification and segmentation of surgical instruments. Instrument segmentation underpins applications such as surgical navigation, skill assessment, and autonomous endoscope guidance (Rueckert et al., 2026), and research interest has grown steadily: a recent bibliometric study of deep learning for instrument segmentation, detection, and tracking in MIS reports sustained growth in annual output between 2017 and 2024, peaking in 2023 (Yousef et al., 2026).

The surgical environment is dynamic and unpredictable. Graspers, scissors, and dissectors appear at varying angles, are partly hidden by tissue, and can be obscured by smoke or blood, while their metallic surfaces create specular highlights. A comparative validation study found that modern deep learning methods outperform other approaches for instrument segmentation, yet the results were still far from perfect and instrument tracking in conventional laparoscopic surgery remained an open challenge (Bodenstedt et al., 2018). Hand-crafted edge detection is therefore not a sufficient basis for clinically useful systems.

This paper argues that laparoscopic instrument segmentation is best treated as a staged problem. Multi-scale features help the model see both large instrument bodies and small tips; a refinement stage sharpens boundaries and class decisions; and a temporal module keeps predictions stable from one frame to the next. Surgical context and VLM-based quality checks are promising additions, but they should be evaluated rather than assumed. The paper makes three contributions: (1) a synthesis of recent evidence on each design element, (2) a conceptual segment-then-refine framework, and (3) a validation plan on public datasets. It is a literature-based study; it does not report new experimental results.

Literature Review

From single-frame CNNs to multi-scale models

Early deep-learning work on instrument segmentation relied on pixel-level classification with convolutional neural networks (CNNs). Because instruments occupy very different fractions of the frame depending on their distance from the camera, later work introduced feature fusion across scales. Wang et al. (2021) proposed a multi-scale and multi-level feature network for instrument segmentation, reflecting the need to combine global context with fine local detail. Angeles-Ceron

et al. (2022) combined attention with multi-scale feature fusion in a lightweight single-stage instance segmentation model evaluated on the ROBUST-MIS 2019 data (more than 10,000 frames), and reported real-time performance above 60 frames per second. Daneshgar Rahbar et al. (2024) took a systems perspective, building a CNN-based real-time instrument segmentation pipeline for *open* surgery and arguing that real-time instrument tracking can streamline surgical workflow. Annotation cost is another constraint: Yang et al. (2022) trained from rough scribble annotations and reported 76.82% IoU and 85.70% Dice for binary instrument segmentation on EndoVis 2017, close to fully supervised performance without requiring pixel-accurate ground truth.

Temporal modeling

Single-frame predictions tend to flicker in video. González et al. (2020) addressed this in ISINet by adding a temporal consistency module to an instance-based segmentation approach, and observed that the module improved performance by better preserving instrument identity across time on EndoVis 2017 and 2018. Zhang et al. (2025) pushed the idea further with an instance-level spatiotemporal design, described next.

Segment-then-classify and VLM-guided refinement

Zhang et al. (2025) proposed a segment-then-classify framework that decouples mask generation from semantic classification. A Mask2Former-based backbone first produces class-agnostic instance masks and region features; a bounding-box-guided module then fuses geometric priors and temporal consistency through a lightweight transformer encoder. The separation is attractive for laparoscopy because localizing an instrument and deciding which instrument it is can fail for different reasons.

Lou et al. (2026), in a recent arXiv preprint, introduced IR-SIS, an iterative refinement system for surgical image segmentation that accepts natural-language descriptions. IR-SIS uses a fine-tuned SAM3 model for initial segmentation, a vision-language model to detect instruments and judge segmentation quality, and an agentic workflow that selects refinement strategies adaptively. Clinicians can also give natural-language feedback during refinement. The authors built a language-annotated dataset from EndoVis 2017 and 2018 and report state-of-the-art results on in-domain and out-of-distribution data. Importantly, the VLM in this design acts as a *quality evaluator* for masks; it is not described as reasoning about surgical phase. As a preprint, its results should be treated as provisional until independently reproduced.

Surgical context and datasets

Rueckert et al. (2026) report the PhaKIR challenge held with EndoVis at MICCAI 2024. They note that incorporating surgical context, such as the current procedural phase, has emerged as a

promising way to improve robustness and interpretability, and they provide a multi-center dataset of thirteen full-length laparoscopic cholecystectomy videos from three institutions, with unified annotations for phase recognition, instrument keypoints, and instrument instance segmentation. This is directly relevant to the present topic because it pairs segmentation with temporal context on laparoscopic data. By contrast, the widely used Cholec80 collection (80 cholecystectomy videos from 13 surgeons) was designed for phase and tool-presence recognition (Twinanda et al., 2017) and does not itself contain pixel-level masks; CholecSeg8k fills that role by providing pixel-level annotation of 13 classes on 8,080 frames drawn from 17 Cholec80 clips (Hong et al., 2020).

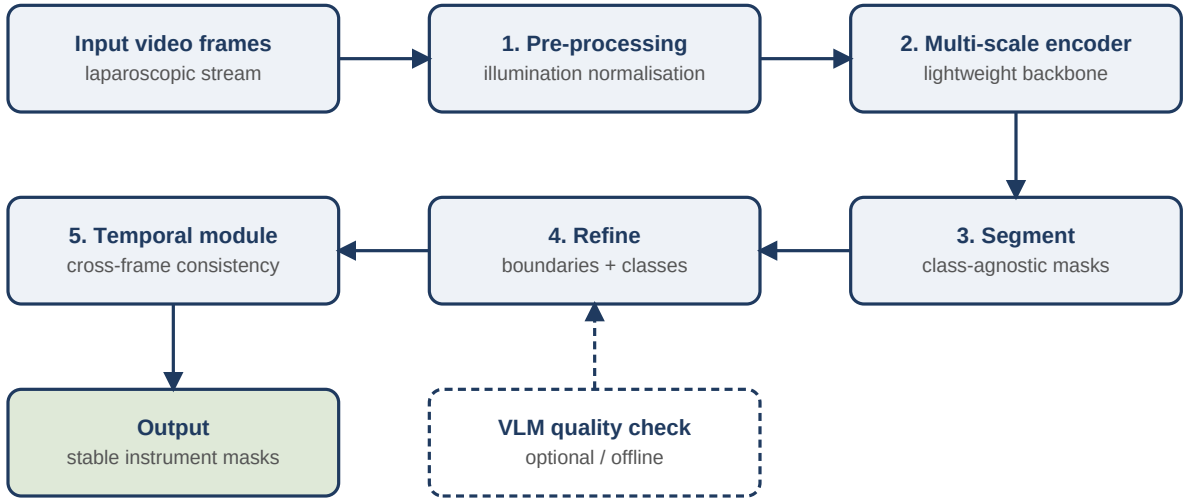
Research gap

Across the sources reviewed here, three gaps stand out. First, multi-scale fusion, refinement, and temporal modeling are usually studied separately, so their combined contribution on laparoscopic cholecystectomy data is not well characterized. Second, robustness is typically assessed through cross-dataset tests rather than through rare, event-level failures such as sudden smoke, equipment malfunction, or unusual anatomy; to the best of this review's knowledge, there is no standard benchmark for such events. Third, the available multi-center laparoscopic data remain small. This paper responds to the first two gaps with a design and an evaluation protocol, and it acknowledges the third as a limit on any conclusions.

Methodology

This section describes the reasoning behind a proposed segment-then-refine pipeline. The pipeline is conceptual: it adapts the decoupling idea from segment-then-classify work (Zhang et al., 2025) and the iterative-refinement idea from VLM-guided work (Lou et al., 2026) to a laparoscopic video setting. Decoupling the initial localization of a tool from the fine-tuning of its boundaries is attractive because reflective metal surfaces make boundaries hard to place in a single pass.

Figure 1



Proposed segment-then-refine pipeline. The dashed VLM quality check is optional and intended for offline review or low-confidence frames, not for the real-time path.

The pipeline has the following stages:

- **Pre-processing.** Video frames are normalized to reduce the effect of the non-uniform lighting produced by laparoscopic light sources.
- **Multi-scale encoder.** A lightweight backbone with a feature pyramid captures instrument signatures at several resolutions, so that thin tips are not lost to aggressive downsampling.
- **Segment.** The first stage produces class-agnostic instance masks that localize each instrument.
- **Refine.** A second stage sharpens boundaries and assigns instrument classes, operating on regions around the predicted instruments.
- **Temporal consistency.** A lightweight temporal module lets the prediction for one frame be informed by neighboring frames, reducing the flicker typical of frame-by-frame analysis.
- **VLM quality check (optional).** A vision-language model reviews selected masks and flags likely errors. Because this adds latency and cost, it is treated as an add-on to be tested, not a requirement of the core system.

Datasets. Cholec80 on its own is not a segmentation benchmark, so the validation plan uses datasets that carry pixel-level labels (Table 1). Using several datasets also makes it possible to study domain shift between robot-assisted and conventional laparoscopic footage.

Table 1

Dataset	Content	Annotation	Intended role
Cholec80 (Twinanda et al., 2017)	80 cholecystectomy videos from 13 surgeons	Phase and tool-presence labels; no pixel masks	Source video only
CholecSeg8k (Hong et al., 2020)	8,080 frames from 17 Cholec80 clips	Pixel-level masks, 13 classes	Primary training and evaluation on laparoscopic data
EndoVis 2017 / 2018	Standard instrument segmentation benchmarks (robot-assisted)	Instrument masks	Comparison with published methods (e.g., González et al., 2020; Yang et al., 2022)
PhaKIR (Rueckert et al., 2026)	13 full-length laparoscopic cholecystectomy videos, 3 institutions	Phase, keypoints, instance masks	Multi-center test and phase-context experiments

Evaluation protocol. Models will be compared using Dice and intersection-over-union (IoU) per instrument class, a boundary-accuracy measure, a temporal stability measure (agreement between masks on consecutive frames), and inference speed in frames per second. Where possible, frames will be tagged by condition (smoke, blood, specular reflection, heavy occlusion) so that performance can be reported per condition rather than only as an overall average.

Safety and limitations. In a live operation, boundary errors could matter, so any real use would require error thresholds set with clinicians and a system designed as decision support rather than autonomous control. This study makes no clinical claims. A further limitation is computational: low-latency processing is demanding, and the proposed use of lightweight backbones is motivated by reports that compact models can reach real-time speeds on benchmark data (Angeles-Ceron et al., 2022), although this has not been verified for the full framework proposed here.

Findings and Discussion

The findings below are a synthesis of evidence from the reviewed literature. They are not results of new experiments, and they are grouped around three sub-claims about the stability and accuracy of segmentation masks.

Sub-claim 1: Temporal context improves stability

Both ISINet (González et al., 2020) and the instance-level spatiotemporal design of Zhang et al. (2025) add explicit cross-frame modeling, and the ISINet authors report that doing so improved performance by preserving instrument identity over time. The practical interpretation is that a temporal module acts like a predictive envelope around known tool locations, damping noise in high-motion sequences. Two cautions apply: temporal modeling adds latency, and it can carry an early error forward across frames, which is why the evaluation plan measures temporal stability

and accuracy together.

Sub-claim 2: Multi-scale fusion helps small and variable-size instruments

The multi-scale network of Wang et al. (2021) and the attention-plus-fusion model of Angeles-Ceron et al. (2022) are consistent with the view that fusing features across resolutions is useful when instruments appear at different sizes. The specific mechanism, that small instrument tips are easily lost in global downsampling, is a reasoned explanation rather than a finding demonstrated by these papers, and it is one of the hypotheses the proposed ablation is designed to test.

Sub-claim 3: Context and VLM guidance are promising but early

PhaKIR frames procedural phase as a promising source of context for robustness and interpretability (Rueckert et al., 2026), and IR-SIS shows that a VLM can be used to detect instruments, assess mask quality, and trigger adaptive refinement, with reported gains on in-domain and out-of-distribution data (Lou et al., 2026). Together they suggest a hybrid direction that combines spatial-temporal features with VLM-based checks. The evidence is still thin: the VLM work is a preprint evaluated on EndoVis-derived data, and neither study establishes how much these additions help in rare, high-occlusion events.

Challenging conditions remain an open question

Smoke from electrocautery, blood, and specular reflection are widely recognized as difficult, and the comparative study by Bodenstedt et al. (2018) shows that the problem was far from solved even for strong deep learning methods. However, the sources reviewed here do not report per-condition results in a way that allows a firm conclusion about, for example, performance during active cauterization. That is why the validation plan includes condition-tagged evaluation: it converts an unsupported intuition into a measurable question.

Dataset and domain considerations

EndoVis 2017 and 2018 are robot-assisted benchmarks, whereas CholecSeg8k and PhaKIR are conventional laparoscopic cholecystectomy data. Methods that perform well on one may not transfer to the other, so results should be reported per dataset. The Daneshgar Rahbar et al. (2024) system targets open surgery with a different camera arrangement and is cited here only as evidence of interest in real-time instrument tracking, not as a laparoscopic baseline.

Planned Experimental Validation

To move from proposal to evidence, the framework will be built incrementally and each component tested in isolation (Table 2). The VLM check is deliberately last because it is the

costliest and least necessary component. Results will be reported per dataset and, where tags are available, per challenging condition.

Table 2

Configuration	Multi-scale	Refinement	Temporal	VLM check
A. Baseline (single-scale, single-frame)	No	No	No	No
B. + Multi-scale fusion	Yes	No	No	No
C. + Refinement stage	Yes	Yes	No	No
D. + Temporal module	Yes	Yes	Yes	No
E. + VLM quality check (optional)	Yes	Yes	Yes	Yes

Planned ablation design. No results are reported in this paper.

Conclusion

This paper reviewed the progression from single-frame CNNs to multi-scale, temporally aware, and VLM-refined approaches to surgical instrument segmentation, and proposed a staged segment-then-refine framework that brings these ideas together for laparoscopic video. The reviewed evidence supports temporal modeling and multi-scale fusion as useful design elements, treats surgical context and VLM guidance as promising but early, and shows that the question of robustness under smoke, blood, and rare events is still open. Reliable instrument segmentation is best understood as a foundational component for safer computer-assisted surgery, not a complete solution on its own.

The study has clear limits. It is a selective literature review rather than a systematic one, some cited work is preprint, no new experiments are reported, and the multi-center laparoscopic data available today, though growing, remain small. Future work should first carry out the validation plan above and then examine cross-domain generalization, for example whether a model trained on gallbladder surgery transfers to other procedures such as thoracic surgery. By connecting raw video to dependable instrument-level information, this line of work can support downstream tasks such as phase recognition and skill assessment, provided it is evaluated with the same rigor as the clinical setting demands.

References

Angeles-Ceron, J. C., Ochoa-Ruiz, G., Chang, L., & Ali, S. (2022). Real-time instance segmentation of surgical instruments using attention and multi-scale feature fusion. *Medical Image Analysis, 81*, 102569. https://doi.org/10.1016/j.media.2022.102569

Bodenstedt, S., et al. (2018). Comparative evaluation of instrument segmentation and tracking methods in minimally invasive surgery. *arXiv*. https://arxiv.org/abs/1805.02475

Daneshgar Rahbar, M., Pappas, G., & Jaber, N. (2024). Toward intraoperative visual intelligence: Real-time surgical instrument segmentation for enhanced surgical monitoring. *Healthcare, 12*(11), 1112. https://doi.org/10.3390/healthcare12111112

González, C., Bravo-Sánchez, L., & Arbeláez, P. (2020). ISINet: An instance-based approach for surgical instrument segmentation. In *Medical Image Computing and Computer-Assisted Intervention (MICCAI 2020)*. Springer. https://arxiv.org/abs/2007.05533

Hong, W.-Y., Kao, C.-L., Kuo, Y.-H., Wang, J.-R., Chang, W.-L., & Shih, C.-S. (2020). CholecSeg8k: A semantic segmentation dataset for laparoscopic cholecystectomy based on Cholec80. *arXiv*. https://arxiv.org/abs/2012.12453

Lou, A., Li, Y., Chang, Q., Xi, N., Xie, L., Li, Z., & Luan, T. (2026). *VLM-guided iterative refinement for surgical image segmentation with foundation models* [Preprint]. arXiv. https://doi.org/10.48550/arXiv.2602.09252

Rueckert, T., Rauber, D., Maerkl, R., Klausmann, L., Yildiran, S. R., Gutbrod, M., Nunes, D. W., Moreno, A. F., Luengo, I., Stoyanov, D., Toussaint, N., Cho, E., Kim, H. B., Choo, O. S., Kim, K. Y., Kim, S. T., Arantes, G., Song, K., Zhu, J., … Palm, C. (2026). Comparative validation of surgical phase recognition, instrument keypoint estimation, and instrument instance segmentation in endoscopy: Results of the PhaKIR 2024 challenge. *Medical Image Analysis*, 103945. https://doi.org/10.1016/j.media.2026.103945

Twinanda, A. P., Shehata, S., Mutter, D., Marescaux, J., de Mathelin, M., & Padoy, N. (2017). EndoNet: A deep architecture for recognition tasks on laparoscopic videos. *IEEE Transactions on Medical Imaging, 36*(1), 86–97. https://doi.org/10.1109/TMI.2016.2593957

Wang, Y., Qiu, Z., Hu, Y., Chen, H., Ye, F., & Liu, J. (2021). Surgical instrument segmentation based on multi-scale and multi-level feature network. In *2021 43rd Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC)* (pp. 2672–2675). IEEE. https://doi.org/10.1109/EMBC46164.2021.9629891

Yang, Z., Simon, R., & Linte, C. A. (2022). A weakly supervised learning approach for surgical instrument segmentation from laparoscopic video sequences. In *Proceedings of SPIE, 12034: Medical Imaging 2022: Image-Guided Procedures, Robotic Interventions, and Modeling*, 120341U. https://doi.org/10.1117/12.2610778

Yousef, M., Aly, K. E., Ahmed, M., Ahmed, F. A., al Jalham, K., & Balakrishnan, S. (2026). Bibliometric analysis of deep learning for surgical instrument segmentation, detection and tracking in minimally invasive surgery. *Frontiers in Digital Health, 8*. https://doi.org/10.3389/fdgth.2026.1633888

Zhang, T., Yuan, X., & Xu, H. (2025). Surgical instrument segmentation via segment-then-classify framework with instance-level spatiotemporal consistency modeling. *Journal of Imaging, 11*(10), 364. https://doi.org/10.3390/jimaging11100364